

1-3 $\mathcal{F}_b = \{A \subset \mathcal{U} : A \text{ is countable or } A^c \text{ is countable}\}$. $\mathcal{U}$ is nonempty set.

i) $\emptyset = \mathcal{U}^c$ is countable $\Rightarrow \emptyset \in \mathcal{F}_b$.

ii) If $A \in \mathcal{F}_b \Rightarrow$ $\begin{cases} \text{case one} = A \text{ is countable} \Rightarrow (A^c)^c \text{ is countable} \Rightarrow A^c \in \mathcal{F}_b \\ \text{case two} = A^c \text{ is countable} \Rightarrow A^c \in \mathcal{F}_b \end{cases}$
 $\Rightarrow A^c \in \mathcal{F}_b$.

iii) $\forall A, B \in \mathcal{F}_b \Rightarrow$ case one: at least one of A and B is countable

$\Rightarrow A \cap B$ is countable

$\Rightarrow A \cap B \in \mathcal{F}_b$.

case two: A^c is countable and B^c is countable.

$\Rightarrow A^c \cup B^c$ is countable

$A \cap B = (A^c \cup B^c)^c$ and $A^c \cup B^c$ is countable

$\Rightarrow A \cap B \in \mathcal{F}_b$.

$\Rightarrow A \cap B \in \mathcal{F}_b$.

So $\mathcal{F}_b$ is an algebra.

iv). $\forall A_n \in \mathcal{F}_b$ s.t. $A_n \subset A_{n+1}$

If $\bigcup_{n=1}^{\infty} A_n$ is countable $\Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}_b$

If $\bigcup_{n=1}^{\infty} A_n$ is not countable $\Rightarrow \exists A_k \in \{A_n\}$ and A_k is uncountable.

But $A_k \in \mathcal{F}_b$
 A_k is uncountable $\Rightarrow A_k^c$ is countable

And $(\bigcup_{n=1}^{\infty} A_n)^c = (\lim_{n \rightarrow \infty} A_n)^c \subset A_k^c$

So $(\bigcup_{n=1}^{\infty} A_n)^c$ is countable

$\Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}_b$

So $\mathcal{F}_b$ is a σ -algebra.

1.8.

$$A_i = \{i, i+1, i+2, \dots\}$$

$$A_{i+1} = \{i+1, i+2, \dots\}$$

$$A_{i+2} = \{i+2, \dots\}$$

first, $\sigma(A) = \bigcap_{F \in \mathcal{F}} F$ and $\mathcal{F}(\mathcal{N})$ is an σ -algebra containing A .

$$\text{So } \sigma(A) \subset \mathcal{F}(\mathcal{N}) \quad (1)$$

$$\text{secondly } \forall i \in \mathbb{N}. \{i\} = A_i \cap A_{i+1}^c$$

$$\forall A \in \mathcal{F}(\mathcal{N}) \quad A = \bigcup_{j \in A} A_j \cap A_{j+1}^c$$

$$\Rightarrow A \in \sigma(A)$$

$$\Rightarrow \mathcal{F}(\mathcal{N}) \subset \sigma(A) \quad (2)$$

$$(1), (2) \Rightarrow \sigma(A) = \mathcal{F}(\mathcal{N}).$$

1.10.

$$\text{start with } k=2. \quad \mathcal{O}_1 = \{(a_1, b_1) \times (a_2, b_2) : -\infty < a_1 < b_1 < +\infty; -\infty < a_2 < b_2 < +\infty\}$$

$$\mathcal{O}_2 = \{(-\infty, x_1) \times (-\infty, x_2) : x_1, x_2 \in \mathbb{R}\}$$

$$\mathcal{O}_3 = \{(a_1, b_1) \times (a_2, b_2) : a_1, a_2, b_1, b_2 \in \mathbb{Q}, a_1 < b_1, a_2 < b_2\}$$

$$\mathcal{O}_4 = \{(-\infty, x_1) \times (-\infty, x_2) : x_1, x_2 \in \mathbb{Q}\}$$

from the example, we know $\mathcal{B}(\mathbb{R}^k) = \sigma(\mathcal{O}_1) = \sigma(\mathcal{O}_3)$

$$\forall (a_1, b_1) \times \dots \times (a_k, b_k) \in \mathcal{O}_3 \quad a_i, b_i \in \mathbb{Q}, \quad a_i \leq b_i \quad (1 \leq i \leq k).$$

$$\text{let } A_i = (-\infty, b_1) \times (-\infty, b_2) \times \dots \times (-\infty, b_{i-1}) \times (a_i, b_i) \times (-\infty, b_{i+1}) \times \dots \times (-\infty, b_k)$$

$$A_1 = (a_1, b_1) \times (-\infty, b_2) \times \dots \times (-\infty, b_k)$$

$$A_k = (-\infty, b_1) \times \dots \times (-\infty, b_{k-1}) \times (a_k, b_k)$$

$$\begin{aligned} \Rightarrow A_i &= \bigcap_{1 \leq j \leq k} A_j = (-\infty, b_1) \times \dots \times (-\infty, b_{i-1}) \times \bigcup_{n=1} \left[(-\infty, b_i) \setminus (-\infty, a_i + \frac{1}{n}) \right] \times (-\infty, b_{i+1}) \times \dots \times (-\infty, b_k) \\ &= \bigcup_{n=1} \left\{ (-\infty, b_1) \times (-\infty, b_2) \times \dots \times (-\infty, b_{i-1}) \times \left[(-\infty, b_i) \setminus (-\infty, a_i + \frac{1}{n}) \right] \times (-\infty, b_{i+1}) \times \dots \times (-\infty, b_k) \right\} \\ &= \bigcup_{n=1} \left\{ (-\infty, b_1) \times \dots \times (-\infty, b_k) \setminus \left((-\infty, b_1) \times \dots \times (-\infty, b_{i-1}) \times (-\infty, a_i + \frac{1}{n}) \times (-\infty, b_{i+1}) \times \dots \times (-\infty, b_k) \right) \right\} \\ &\in \sigma(\mathcal{O}_4) \end{aligned}$$

$$\text{So } \bigcap_{1 \leq i \leq k} A_i \in \sigma(\mathcal{O}_4) \Rightarrow \mathcal{O}_3 \subset \sigma(\mathcal{O}_4)$$

1.18

$$\mathcal{N} = \mathcal{N}, \quad \mathcal{F} = \mathcal{F}(\mathcal{N}), \quad A_n = \{n, n+1, \dots\}$$

$$\mu(A_n) = \infty \text{ for all } n \in \mathcal{N} \Rightarrow \lim_{n \rightarrow \infty} \mu(A_n) = \infty.$$

$$A_n \cap A_{n+1} = A_{n+1} \Rightarrow \mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \mu(\emptyset) = 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} \mu(A_n) \neq \mu\left(\bigcap_{n=1}^{\infty} A_n\right)$$

1.19. $\mathcal{L} \subset \mathcal{F}(\mathcal{N})$, a semi-algebra; let $\mathcal{A}(\mathcal{L}) \stackrel{\text{algebra}}{=} \{A = A = \bigcup_{i=1}^k B_i = B_i \in \mathcal{L}, i=1, \dots, k\}$

(a) for any algebra $\mathcal{E}$ containing $\mathcal{L}$. $\Rightarrow \mathcal{A}(\mathcal{L}) \subset \mathcal{E}$.

$$\forall A \in \mathcal{L}, \exists B_1, \dots, B_m \in \mathcal{L} \text{ s.t. } A^c = \bigcup_{i=1}^m B_i$$

$$\text{So } \mathcal{N} = A \cup A^c = A \cup \left[\bigcup_{i=1}^m B_i\right] \in \mathcal{A}(\mathcal{L}).$$

$$\text{Also, since } A = \bigcup_{i=1}^k B_i = B_i \in \mathcal{L}, \text{ so if } A_1, \dots, A_n, \dots \in \mathcal{A}(\mathcal{L}) \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{A}(\mathcal{L})$$

$$\text{for any set } A \in \mathcal{A}(\mathcal{L}), A = \bigcup_{i=1}^k B_i, B_i \in \mathcal{L}, i \in \mathcal{N}.$$

$$\text{Similarly, } B_i^c = \bigcup_{j=1}^{n_i} B_{ij}^c, \quad A^c = \bigcap_{i=1}^k B_i^c = \bigcap_{i=1}^k \left[\bigcup_{j=1}^{n_i} B_{ij}^c \right]$$

$$\Rightarrow A^c = \bigcap_{i=1}^k \left[\bigcup_{j=1}^{n_i} B_{ij}^c \right] = \bigcup_{j=1}^{n_1} \dots \bigcup_{j=k}^{n_k} \left(\bigcap_{m=1}^k B_{mj}^c \right) \in \mathcal{A}(\mathcal{L})$$

$\Rightarrow \mathcal{A}(\mathcal{L})$ is the smallest algebra generated by $\mathcal{L}$.

$$(b). \mathcal{L} \subset \mathcal{A}(\mathcal{L}) \Rightarrow \sigma(\mathcal{L}) \subset \sigma(\mathcal{A}(\mathcal{L})), \mathcal{A}(\mathcal{L}) \subset \sigma(\mathcal{L})$$

$$\Rightarrow \sigma(\mathcal{A}(\mathcal{L})) \subset \sigma(\sigma(\mathcal{L})) = \sigma(\mathcal{L})$$

$$\Rightarrow \sigma(\mathcal{L}) = \sigma(\mathcal{A}(\mathcal{L}))$$

1.20.

$$(3.4) \quad 0 \leq \mu^*(\emptyset) \leq \sum_{n=1}^{\infty} \mu(\emptyset) = 0 \Rightarrow \mu^*(\emptyset) = 0$$

$$(3.5) \quad A \subset B \Rightarrow \Sigma(B) \subset \Sigma(A) \Rightarrow \mu^*(A) = \inf \Sigma(A) \leq \inf \Sigma(B) = \mu^*(B)$$

$$(3.6) \quad \forall \varepsilon > 0, \text{ for every } i, \exists \{A_{ij}\}_{j=1}^{\infty} \subset \mathcal{C}, A_i \subset \bigcup_{j=1}^{\infty} A_{ij}, \text{ s.t. } \sum_{j=1}^{\infty} \mu(A_{ij}) \leq \mu^*(A_i) + \frac{\varepsilon}{2^i}$$

$$\Rightarrow \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \mu(A_{ij}) \leq \sum_{i=1}^{\infty} \mu^*(A_i) + \varepsilon.$$

$$\Rightarrow \bigcup_{i=1}^{\infty} A_i \subset \bigcup_{i=1}^{\infty} \bigcup_{j=1}^{\infty} A_{ij}, \quad \{A_{ij} = i \geq 1, j \geq 1\} \subset \mathcal{C}$$

$$\Rightarrow \mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \mu(A_{ij}) \leq \sum_{i=1}^{\infty} \mu^*(A_i) + \varepsilon.$$

$$\forall \varepsilon > 0 \Rightarrow \mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$$

1.24.

$$C = \{ (a, b] : -\infty \leq a \leq b < +\infty \} \cup \{ (a, +\infty) : -\infty \leq a < +\infty \}$$

$$C_2 = \{ I_1 \times I_2 : I_1, I_2 \in C \}$$

$$\forall I_1 \times I_2, I_3 \times I_4 \in C_2, I_1 \times I_2 \cap I_3 \times I_4 = (I_1 \cap I_3) \times (I_2 \cap I_4) \in C_2.$$

$$\forall I_1 \times I_2 \in C_2, (I_1 \times I_2)^c = (I_1^c \times I_2) \cup (I_1 \times I_2^c) \cup (I_1^c \times I_2^c)$$

since $I_1, I_2 \in C$,

$$I_1^c = \bigcup_{j=1}^{n_1} I_{1j}, I_{1j_1} \cap I_{1j_2} = \emptyset, j_1 \neq j_2, I_{11}, \dots, I_{1n_1} \in C.$$

$$\text{Similarly } I_2^c = \bigcup_{j=1}^{n_2} I_{2j}, I_{2j_1} \cap I_{2j_2} = \emptyset, j_1 \neq j_2, I_{21}, \dots, I_{2n_2} \in C$$

$$I_1 \times I_2^c \cap I_1^c \times I_2 = \emptyset$$

$$I_1 \times I_2^c \cap I_1^c \times I_2^c = \emptyset$$

$$I_1^c \times I_2 \cap I_1^c \times I_2^c = \emptyset$$

$$I_{1j} \cap I_{1l} = \emptyset, j \neq l$$

$$I_{2j} \cap I_{2l} = \emptyset, j \neq l$$

and

$$I_1^c \times I_2 = \bigcup_{j=1}^{n_1} (I_{1j} \times I_2)$$

$$I_1 \times I_2^c = \bigcup_{j=1}^{n_2} (I_1 \times I_{2j})$$

$$I_1^c \times I_2^c = \bigcup_{j=1}^{n_1} \bigcup_{l=1}^{n_2} (I_{1j} \times I_{2l})$$

$\Rightarrow (I_1 \times I_2)^c$ is a finite union of disjoint sets from C_2 .

$\Rightarrow (I_1 \times I_2)^c \in C_2$.

Extra Problem:

$$(i) \quad \forall x_1 < x_2 \in \mathbb{R}.$$

$$\text{if } x_1 < x_2 \leq 0 \quad F(x_1) = F(x_2) = 0 + 0 = 0$$

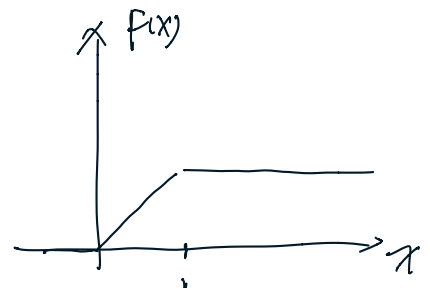
$$\text{if } x_1 \leq 0 < x_2 < 1 \quad F(x_1) = 0 \leq x_2 = F(x_2)$$

$$\text{if } 0 < x_1 < x_2 < 1 \quad F(x_1) = x_1 < x_2 = F(x_2)$$

$$\text{if } 0 < x_1 < 1 < x_2 \quad F(x_1) = x_1 < 1 = F(x_2)$$

$$\text{if } 1 \leq x_1 < x_2 \quad F(x_1) = F(x_2) = 1$$

So $F(x)$ is non-decreasing.



obviously $f(x)$ is continuous on $(0,1)$

$$\forall \varepsilon > 0. \quad 0 = f(-\varepsilon) = f(0) = \lim_{\varepsilon \rightarrow 0^+} \varepsilon = \lim_{\varepsilon \rightarrow 0^+} f(\varepsilon).$$

$$1 = f(1+\varepsilon) = f(1) = \lim_{\varepsilon \rightarrow 0^+} f(1-\varepsilon),$$

So $f(x)$ is also continuous.

$$(ii) \quad C \equiv \left\{ (a,b] : -\infty \leq a \leq b < +\infty \right\} \cup \left\{ (a,+\infty) : -\infty \leq a < +\infty \right\}.$$

$$\mu_f((a,b]) = f(b) - f(a)$$

$$\mu_f((a,+\infty)) = f(+\infty) - f(a)$$

$$(iii) \quad \forall y \in \mathbb{R}, \quad x \in (\mathbb{R} \cup \{+\infty\}).$$

$$\Rightarrow (-\infty, y) = \bigcup_{n=1}^{\infty} (-\infty, \frac{y+1}{n}] \Rightarrow O_2 \subset \sigma(C)$$

$$\Rightarrow \mathcal{B}(\mathbb{R}) = \sigma(O_2) \subset \sigma(C)$$

$$(x, y] = \bigcap_{n=1}^{\infty} (x, \frac{y+1}{n}) \Rightarrow C \subset \sigma(O_1)$$

$$\Rightarrow \sigma(C) \subset \mathcal{B}(\mathbb{R})$$

$$\text{So } \sigma(C) = \mathcal{B}(\mathbb{R})$$

(iv) there exists a unique measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ extending μ_f .

(v) By Carathéodory's extension theorem and uniqueness of measures, there exists such μ .